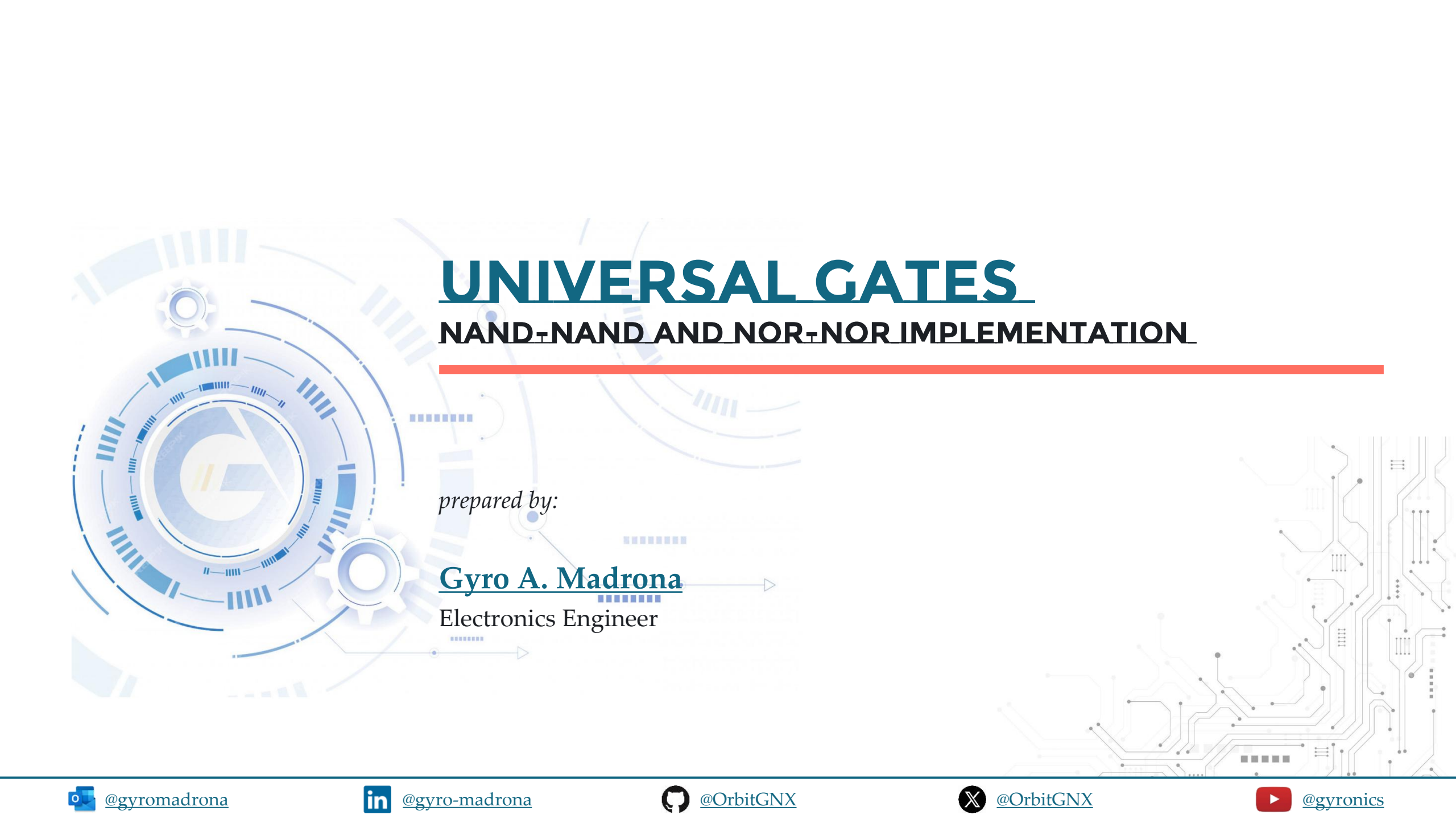




UNIVERSAL GATES

NAND-NAND AND NOR-NOR IMPLEMENTATION

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TOPIC OUTLINE

NAND-NAND Implementation

NOR-NOR Implementation



NAND-NAND IMPLEMENTATION



INVERTER

NAND gate as an Inverter

Solution



AND GATE

NAND gate as an AND gate

Solution



OR GATE

NAND gate as an OR gate

Solution



XOR_GATE

NAND-NAND implementation of an SOP expression

Solution



EXERCISE

Convert the given SOP (Sum of Products) expression into its equivalent NAND-NAND implementation.

Solution

a. $Y = AB + CD$

b. $Y = ABC + DE + FG$

c. $Y = \overline{A}B + C\overline{D} + EF$



NOR-NOR IMPLEMENTATION



INVERTER

NOR gate as an Inverter

Solution



AND_GATE

NOR gate as an AND gate

Solution



OR GATE

NOR gate as an OR gate

Solution



XOR_GATE

NOR-NOR implementation of an SOP expression

Solution



EXERCISE

Convert the given POS (Product of Sum) expression into its equivalent NOR-NOR implementation.

Solution

a. $Y = (A + B)(C + D)$

b. $Y = (A + B + C)(D + E)$

c. $Y = (\bar{A} + B)(C + \bar{D})(E + F)$



LABORATORY

